

----- Active space -----

GROWTH HORMONE DEFICIENCY AND HYPOTHYROIDISM

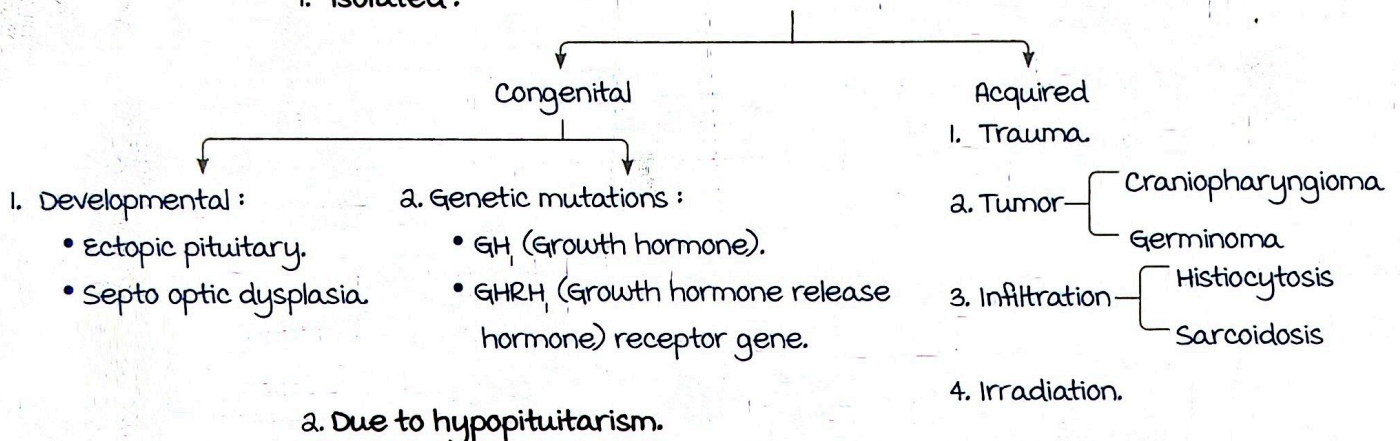
Growth Hormone Deficiency

00:00:17

TYPES :

Overall m/c cause : **Idiopathic**.

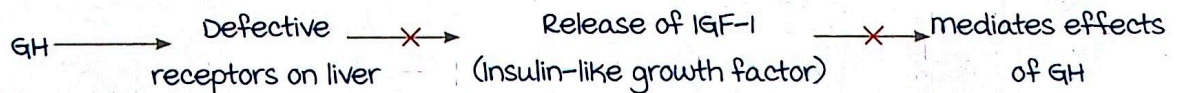
1. Isolated :



LARON'S DWARFISM :

- Due to defective GH receptor (Resistance to effect of GH). khanirfan0392@gmail.com

Pathogenesis :



- **Characteristic feature** : Low levels of IGF₁.

FEATURES :

Isolated GH deficiency :

1. Normal at birth (D/t no role of GH during fetal life).
2. **Proportionate short stature** : During infancy
 - Growth velocity ↓ : < 4 cm/yr
 - mid parental height (MPH) : < -2 SD from MPH
3. **Doll-like appearance** : ↓ GH → ↓ lipolysis → Deposition of fat
 - Prominent cheeks
 - Overweight/obese
4. Bone age : Delayed.

Note :

mid parental height : Genetic potential for height of the child.

If a/w hypopituitarism:

----- Active space -----

1. ↓ gonadotropins → ↓ testosterone & other sex hormones { micropenis & small testes
Delayed puberty
2. ↓ ACTH → ↓ cortisol: Hypoglycemia.
3. Prolonged jaundice.

INVESTIGATIONS:

Screening:

1. IGF-1 levels:
 - malnutrition can alter IGF levels.
2. IGF BP-3 (Binding protein) levels:
 - Preferred test when associated with severe malnutrition (+).

Confirmatory:

- GH estimation: Stimulative/provocative GH testing.
 - Random estimation unreliable as GH released in pulsatile manner.
 - **Provocative substances**: Insulin, clonidine, arginine.
 - GH < 10 ng/ml after test: Diagnostic.

TREATMENT

rhGH (Recombinant GH):

- Subcutaneous injection (usually given at night) until epiphyseal closure.

Note:

- Other uses of rhGH:

GTCS IN prader willi syndrome.

G: GH deficiency.

I: Idiopathic.

T: Turner syndrome.

N: Noonan syndrome.

C: CKD (Chronic kidney disease).

P: Prader willi syndrome.

S: SGA (Small for gestational age) > 2 years, **SHOX gene mutation**.

↓
Show catch up growth within 2 years of age

khaniirfan0392@gmail.com

Hypothyroidism

00:15:36

- Importance of early detection (First 2 years): Thyroid is integral in brain development.
- Low IQ if undetected and untreated.

Feedback

CONGENITAL HYPOTHYROIDISM:

m/c preventable cause of intellectual disability.

----- Active space -----

Classification :

Based on pathology :

Primary : Pathology in thyroid gland > Central : Pathology in CNS — { Hypothalamus
Pituitary

Etiology :

1. Anomaly : Thyroid dysgenesis — { Aplasia
Hypoplasia
Ectopic thyroid (m/c : Lingual)
 - m/c cause overall.
 - Sporadic.
2. Thyroid dysmorphogenesis : Defects in organification (m/c) → Thyroid peroxidase deficiency.
 - Inheritance : Autosomal recessive.
3. Severe Iodine deficiency : Endemic goitre/cretinism.
 - Seen in underdeveloped countries.

4. Maternal causes — { Transplacental transfer of maternal antibodies : TSH receptor blocking antibody
Antithyroid drugs

Features :

Difficult to identify in early newborn period (1st week) : D/t masking by maternal thyroid hormones.

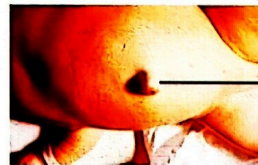
Features > 1 week :

1. ↓ activity of the child.
2. Wide open posterior fontanelle : > 5 mm.
3. Cold dry skin
hypothermia.
4. Constipation/delay in passage of



5. macroglossia :

- ↑ mucopolysaccharides
→ Deposited in the tongue



6. Umbilical hernia

meconium.

7. Hoarse cry.
8. Prolonged jaundice.
9. Goitre is usually absent : In thyroid dysgenesis.
 - Can be present : In dysmorphogenesis & maternal antithyroid drug usage.

Investigations :

1. Thyroid Function Test (TFT) : ↓ T₃, ↓ free T₄, ↑ TSH.
2. Screening (Infants) : Done in all infants.
 - Heel prick dried blood spot testing.

- $\uparrow$ TSH : 2 - 4 days after birth.

↓
If $> 40 \text{ mU/l}$ — Send TFT
Start treatment stat

----- Active space -----

3. Radiological assessment :



Epiphyseal stippling

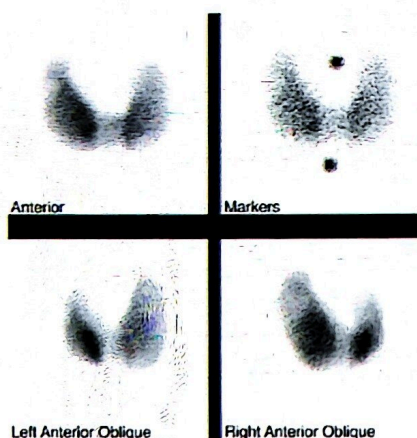
- Delayed bone age.
 - Epiphyseal stippling/dysgenesis (Dot like structures at epiphysis).
- Not diagnostic of hypothyroidism

Confirmatory tests :

- USG of thyroid : Present/absent/ectopic.
- Radionuclide study :
 - Radioisotope used : ^{123}I >> ^{99}Tc .

Treatment :

- Oral thyroxine (T_4) :
 - To be started as early as possible.
 - Dose : 10 - 15 $\mu\text{g/kg/day}$ (High dose to promote brain growth).
 - Goal : Normalization of serum free T_4 .
 - Lifelong treatment.



Radionuclide study of thyroid

ENDEMIC CRETINISM :

- Endemic to countries with severe iodine deficiency.

----- Active space -----

Types :**I. Neurological :**

- most severe.
- very low IQ.
- Goitre +.

Neurological manifestations

- Spasticity
- Squint/nystagmus
- Deaf mutism : D/t to SNHL (Sensorineural hearing loss)

a. myxedematous :

- Low IQ.
- Loss of lateral 1/3rd of eyebrow.
- Periorbital puffiness of facial edema.
- Goitre is usually absent.

ACQUIRED HYPOTHYROIDISM :

- Less common.
- Cause : Hashimoto's thyroiditis → Autoimmune antibody mediated damage.
- Age and gender : Adolescent females.

Features :

- First sign : Growth retardation/ short stature.
- Constipation.
- Dry skin.
- Uncontrollable weight gain despite having poor appetite.
- Goitre + : Firm, nodular swelling.
- Delayed puberty.

648c85cfee3b03a74e182fab

management :

- TFT : ↓ T₃, ↓ T₄, ↑ TSH.
- Autoantibodies : — Anti - TPO (Thyroid peroxidase)
Anti thyroglobulin

Treatment :

- Oral thyroxine (T₄) :
- Dose : 3- 5 μg/ kg/ day (For adequate dosage : Titrate with TSH levels).
- Lifelong treatment.